

Synthetic Vineyard Field Trial Report

Deficit irrigation, soil moisture, grapevine water stress, and disease monitoring

SYNTHETIC-DATA NOTICE: This report was generated solely for AgroAnnotator testing. The vineyard, experimental design, measurements, results, and recommendations are fictional and must not be used for agricultural decisions.

Document type	Fictional field-trial report
Fictional site	Stone Ridge Research Vineyard
Crop	Wine grapevine (Vitis vinifera)
Annotation emphasis	PDF text extraction, repeated concepts, tables, negation, and uncertainty

1. Abstract

This synthetic field trial compares three irrigation strategies in a wine grapevine vineyard: full irrigation, regulated deficit irrigation, and a rainfed treatment. The report deliberately repeats grapevine, vineyard, deficit irrigation, soil moisture, drought stress, evapotranspiration, cover crop, powdery mildew, downy mildew, and Botrytis bunch rot. The goal is to test whether AgroAnnotator extracts text from a PDF, annotates agricultural concepts across pages, merges repeated concepts, and preserves occurrence information.

Fictional observations suggested that regulated deficit irrigation reduced water use while maintaining canopy function during moderate drought. The trial did not confirm powdery mildew, downy mildew, or Botrytis bunch rot. These plant diseases are nevertheless mentioned as monitoring targets, creating useful cases in which a concept is present in the text but the corresponding condition is explicitly absent or uncertain.

2. Background

Water availability influences grapevine growth, berry development, yield, and grape composition. Vineyard irrigation decisions may use rainfall, soil moisture, reference evapotranspiration, crop coefficient, canopy temperature, stem water potential, and visible drought stress. No single measurement is sufficient in every soil and growth stage, so the fictional trial combines sensor data with direct grapevine observations.

The Stone Ridge Research Vineyard is described as a Mediterranean site with warm, dry summers and winter rainfall. The fictional soil is a gravelly clay loam with moderate soil organic matter and uneven water-holding capacity. A winter cover crop of barley and subterranean clover grows between vine rows. The cover crop is terminated before summer to reduce competition for soil moisture, while a low vegetation strip remains to protect soil from erosion.

3. Synthetic experimental design

The fictional experiment contains twelve plots arranged in four blocks. Each block contains one full-irrigation plot, one regulated-deficit-irrigation plot, and one rainfed plot. Every plot contains mature wine grapevine plants of the same cultivar and rootstock. Border vines are excluded from measurements. Drip irrigation is delivered through pressure-compensating emitters.

Treatment	Water-management description	Repeated annotation signals
Full irrigation	Irrigation replaces most estimated crop evapotranspiration after accounting for effective rainfall.	irrigation, evapotranspiration, rainfall
Regulated deficit irrigation	Water supply is reduced during a defined berry-development period and restored if grapevine water stress exceeds a fictional safety limit.	deficit irrigation, grapevine, water stress
Rainfed	No routine irrigation; emergency water would be allowed only to protect permanent vines from severe drought.	rainfed agriculture, drought, soil moisture

4. Measurements

4.1 Weather and water balance

A fictional weather station records rainfall, air temperature, relative humidity, wind speed, solar radiation, and reference evapotranspiration. Irrigation flow meters record the water volume supplied to each treatment. Soil moisture sensors are placed in the grapevine root zone at two depths. Sensor readings are checked against manual soil inspection because stones and roots can affect local soil moisture measurements.

4.2 Grapevine water status

Predawn leaf water potential and midday stem water potential are measured on selected grapevine leaves. Canopy temperature is recorded with a thermal sensor, and stomatal conductance is measured on clear days. Scouts also record leaf angle, shoot growth, tendril condition, leaf yellowing, and berry shrivelling as possible signs of drought stress.

4.3 Yield and fruit quality

At harvest, the fictional team records grape yield, cluster number, berry mass, soluble solids, titratable acidity, pH, and visual fruit condition. A sample of grape berries is retained for repeat analysis. The report uses these measurements as synthetic text only; it does not claim a real relationship between deficit irrigation and wine quality.

5. Pest and disease monitoring

Vineyard scouts inspect grapevine leaves and clusters for powdery mildew, downy mildew, Botrytis bunch rot, grapevine moth, leafhopper, and mite injury. Pheromone traps are used for grapevine moth monitoring. Beneficial organisms such as predatory mites, ladybirds, lacewings, spiders, and parasitoid wasps are also recorded.

No powdery mildew colony was confirmed in the fictional plots. Two pale leaf patches were observed, but they lacked diagnostic fungal growth. Downy mildew was not confirmed. Botrytis bunch rot was also not confirmed at the time of inspection. These negative statements are intentional: AgroAnnotator may detect powdery mildew, downy mildew, and Botrytis bunch rot even though the document states that they were absent.

Disease risk was discussed in relation to canopy density, relative humidity, rainfall, leaf wetness, and berry damage. The canopy was managed by shoot positioning and selective leaf removal. The synthetic report does not prescribe fungicide or insecticide use. Any real plant protection decision would require diagnosis, local authorization, resistance management, crop stage, and professional advice.

6. Synthetic observations

Variable	Full irrigation	Deficit irrigation	Rainfed
Soil moisture	Stable in upper root zone	Moderate seasonal decline	Low during warm period
Grapevine water stress	Low	Moderate and reversible	Visible on two afternoons
Canopy growth	Vigorous	Balanced	Reduced shoot growth
Disease status	No disease confirmed	No disease confirmed	No disease confirmed
Interpretation	Higher water use	Potential water saving	Greatest drought exposure

The table contains qualitative synthetic observations rather than experimental evidence. Repeated phrases in the table help test whether PDF table text appears in the extracted document and whether concepts originating in different PDF elements receive valid global offsets.

7. Interpretation

In the fictional data, regulated deficit irrigation used less water than full irrigation and produced less visible drought stress than the rainfed treatment. Soil moisture declined during the deficit period and recovered after irrigation resumed. These statements are internally consistent test prose, not a conclusion about a real vineyard.

The rainfed grapevine plots showed temporary leaf-angle changes and reduced shoot growth during warm weather. The text mentions severe drought only as an emergency condition that would trigger protective irrigation; severe drought was not observed. This distinction allows evaluation of whether annotations preserve the surrounding context.

Cover crop residue remained between vineyard rows after termination. The cover crop supported soil protection and biological activity but could compete for soil moisture if maintained too long. The fictional team therefore considered cover crop biomass, rainfall, soil type, erosion risk, and grapevine water status together.

8. Recommended verification checks

- 1 Confirm that the PDF is detected as a text-based PDF and that selectable text is extracted from every page.
- 2 Verify that grapevine, vineyard, deficit irrigation, soil moisture, drought stress, evapotranspiration, and cover crop are detected where supported by the selected ontology.
- 3 Check that repeated concepts are merged at document level while retaining multiple occurrences and example phrases.
- 4 Locate annotations originating in a PDF table and confirm that their global character offsets match the extracted text.

- 5 Confirm that disease mentions are not automatically interpreted as diagnoses: powdery mildew, downy mildew, and Botrytis bunch rot were not confirmed.
- 6 Compare the JSON and CSV concept summaries for consistent ontology acronyms, concept URIs, preferred labels, counts, and matched phrases.

9. Limitations of semantic annotation

Ontology annotation identifies concepts that are mentioned; it does not verify the fictional trial design, infer causality, or judge agronomic suitability. A concept match for fungicide does not mean that a fungicide was used. A match for severe drought does not mean that severe drought occurred. A match for Botrytis bunch rot does not mean that disease was diagnosed.

For downstream use, concept annotations should remain linked to the source text so that negation, uncertainty, treatment assignment, measurement context, and time can be examined. The final repeated signals are grapevine, vineyard, regulated deficit irrigation, drip irrigation, soil moisture, drought stress, rainfall, evapotranspiration, cover crop, powdery mildew, downy mildew, Botrytis bunch rot, grapevine moth, and biological control.

End of synthetic PDF fixture. No real vineyard, experiment, or recommendation is represented.